Software Engineering
UE24CS341A

Lab_07: Fuzzing for Input Validation Bugs

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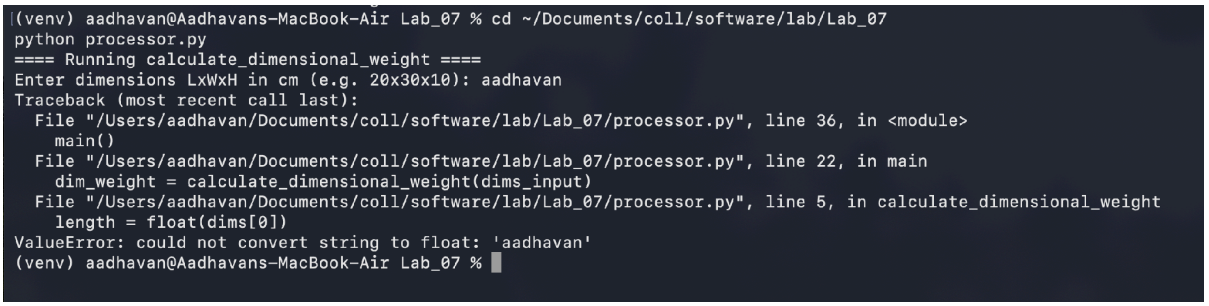
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Submission Repository:
https://github.com/paya5am/Software-Engineering-Lab

Buggy Output

Running python processor.py with the invalid input "aadhavan" crashes with a ValueError.



```
(venv) aadhavan@Aadhavans-MacBook-Air Lab_07 % cd ~/Documents/coll/software/lab/Lab_07
python processor.py
==== Running calculate_dimensional_weight ====
Enter dimensions LxWxH in cm (e.g. 20x30x10): aadhavan
Traceback (most recent call last):
  File "/Users/aadhavan/Documents/coll/software/lab/Lab_07/processor.py", line 36, in <module>
    main()
  File "/Users/aadhavan/Documents/coll/software/lab/Lab_07/processor.py", line 22, in main
    dim_weight = calculate_dimensional_weight(dims_input)
  File "/Users/aadhavan/Documents/coll/software/lab/Lab_07/processor.py", line 5, in calculate_dimensional_weight
    length = float(dims[0])
ValueError: could not convert string to float: 'aadhavan'
(venv) aadhavan@Aadhavans-MacBook-Air Lab_07 %
```

Failing Tests

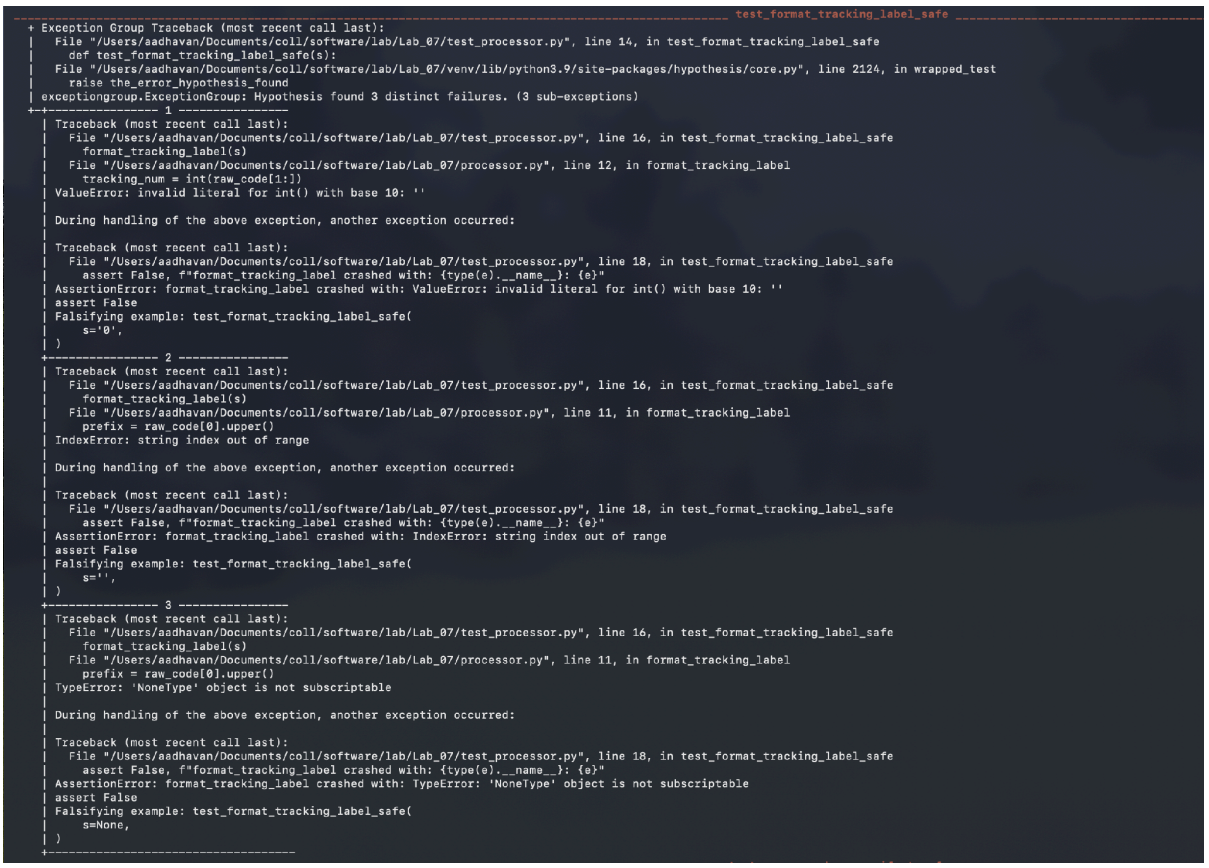
Running python -m pytest test_processor.py against the buggy processor.py: all 3 tests fail.



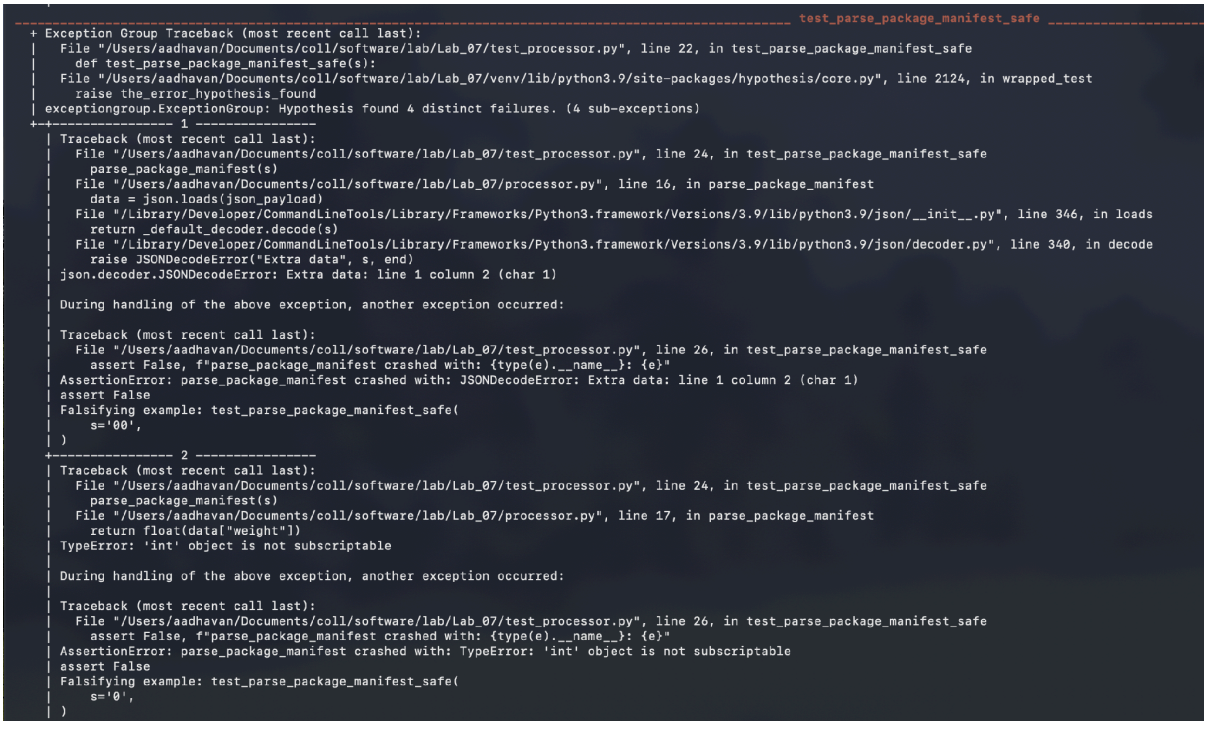
```
(venv) aadhavan@Aadhavans-MacBook-Air Lab_07 % python -m pytest test_processor.py
============================= test session starts ==============================
platform darwin -- Python 3.9.6, pytest-8.4.2, pluggy-1.6.0
rootdir: /Users/aadhavan/Documents/coll/software/lab/Lab_07
plugins: hypothesis-6.141.1
collected 3 items

test_processor.py FFF

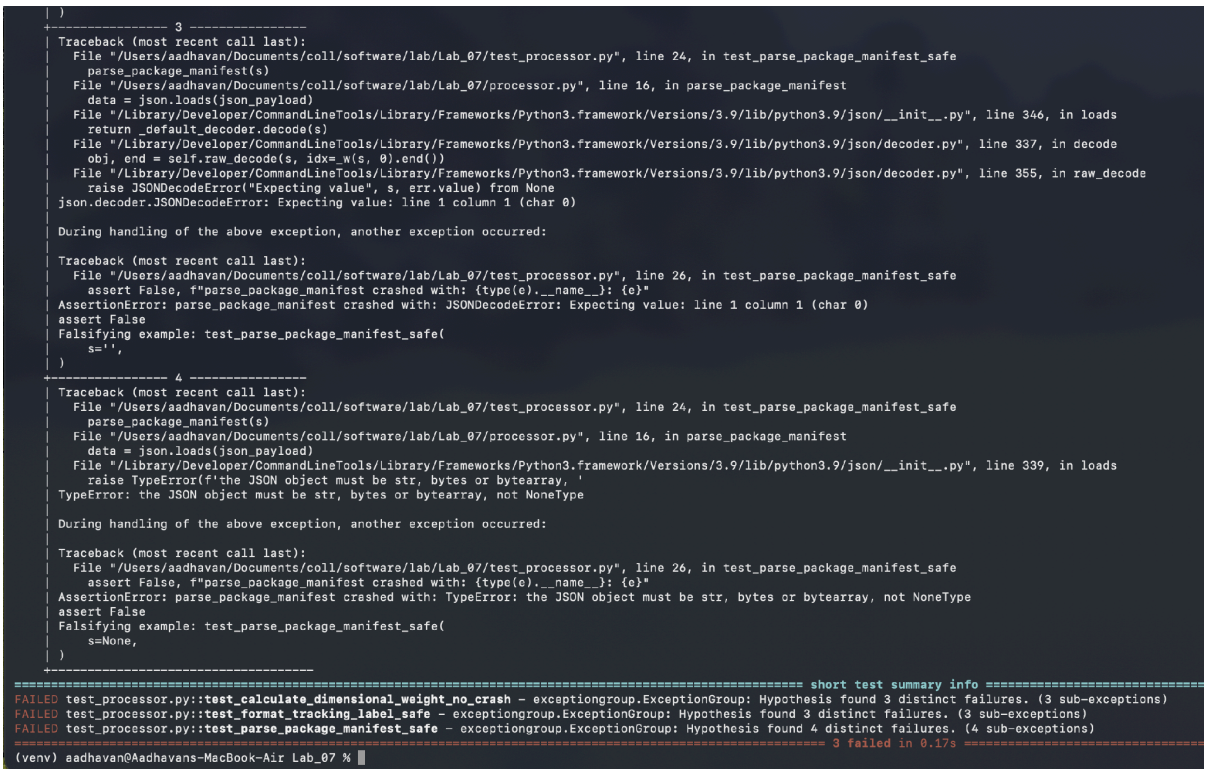
=================================== FAILURES ===================================
_____________________ test_calculate_dimensional_weight_no_crash _____________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/aadhavan/Documents/coll/software/lab/Lab_07/test_processor.py", line 6, in test_calculate_dimensional_weight_no_crash
  |     def test_calculate_dimensional_weight_no_crash(s):
  |   File "/Users/aadhavan/Documents/coll/software/lab/Lab_07/venv/lib/python3.9/site-packages/hypothesis/core.py", line 2124, in wrapped_test
  |     raise the_error_hypothesis_found
  | exceptiongroup.ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/aadhavan/Documents/coll/software/lab/Lab_07/test_processor.py", line 8, in test_calculate_dimensional_weight_no_crash
    |     calculate_dimensional_weight(s)
    |   File "/Users/aadhavan/Documents/coll/software/lab/Lab_07/processor.py", line 6, in calculate_dimensional_weight
    |     width = float(dims[1])
    | IndexError: list index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/aadhavan/Documents/coll/software/lab/Lab_07/test_processor.py", line 10, in test_calculate_dimensional_weight_no_crash
    |     assert False, f"calculate_dimensional_weight crashed with: {type(e).__name__}: {e}"
    | AssertionError: calculate_dimensional_weight crashed with: IndexError: list index out of range
    | assert False
    | Falsifying example: test_calculate_dimensional_weight_no_crash(
    |     s='0',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/aadhavan/Documents/coll/software/lab/Lab_07/test_processor.py", line 8, in test_calculate_dimensional_weight_no_crash
    |     calculate_dimensional_weight(s)
    |   File "/Users/aadhavan/Documents/coll/software/lab/Lab_07/processor.py", line 5, in calculate_dimensional_weight
    |     length = float(dims[0])
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/aadhavan/Documents/coll/software/lab/Lab_07/test_processor.py", line 10, in test_calculate_dimensional_weight_no_crash
    |     assert False, f"calculate_dimensional_weight crashed with: {type(e).__name__}: {e}"
    | AssertionError: calculate_dimensional_weight crashed with: ValueError: could not convert string to float: ''
    | assert False
    | Falsifying example: test_calculate_dimensional_weight_no_crash(
    |     s='',
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "/Users/aadhavan/Documents/coll/software/lab/Lab_07/test_processor.py", line 8, in test_calculate_dimensional_weight_no_crash
    |     calculate_dimensional_weight(s)
    |   File "/Users/aadhavan/Documents/coll/software/lab/Lab_07/processor.py", line 4, in calculate_dimensional_weight
    |     dims = dimensions_str.split('x')
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/aadhavan/Documents/coll/software/lab/Lab_07/test_processor.py", line 10, in test_calculate_dimensional_weight_no_crash
    |     assert False, f"calculate_dimensional_weight crashed with: {type(e).__name__}: {e}"
    | AssertionError: calculate_dimensional_weight crashed with: AttributeError: 'NoneType' object has no attribute 'split'
    | assert False
    | Falsifying example: test_calculate_dimensional_weight_no_crash(
    |     s=None,
    | )
    +------------------------------------
```



```
_____ test_format_tracking_label_safe _____
  + Exception Group Traceback (most recent call last):
  |   File "/Users/aadhavan/Documents/coll/software/lab/Lab_07/test_processor.py", line 14, in test_format_tracking_label_safe
  |     def test_format_tracking_label_safe(s):
  |   File "/Users/aadhavan/Documents/coll/software/lab/Lab_07/venv/lib/python3.9/site-packages/hypothesis/core.py", line 2124, in wrapped_test
  |     raise the_error_hypothesis_found
  | exceptiongroup.ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/aadhavan/Documents/coll/software/lab/Lab_07/test_processor.py", line 16, in test_format_tracking_label_safe
    |     format_tracking_label(s)
    |   File "/Users/aadhavan/Documents/coll/software/lab/Lab_07/processor.py", line 12, in format_tracking_label
    |     tracking_num = int(raw_code[1:])
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/aadhavan/Documents/coll/software/lab/Lab_07/test_processor.py", line 18, in test_format_tracking_label_safe
    |     assert False, f"format_tracking_label crashed with: {type(e).__name__}: {e}"
    | AssertionError: format_tracking_label crashed with: ValueError: invalid literal for int() with base 10: ''
    | assert False
    | Falsifying example: test_format_tracking_label_safe(
    |     s='0',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/aadhavan/Documents/coll/software/lab/Lab_07/test_processor.py", line 16, in test_format_tracking_label_safe
    |     format_tracking_label(s)
    |   File "/Users/aadhavan/Documents/coll/software/lab/Lab_07/processor.py", line 11, in format_tracking_label
    |     prefix = raw_code[0].upper()
    | IndexError: string index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/aadhavan/Documents/coll/software/lab/Lab_07/test_processor.py", line 18, in test_format_tracking_label_safe
    |     assert False, f"format_tracking_label crashed with: {type(e).__name__}: {e}"
    | AssertionError: format_tracking_label crashed with: IndexError: string index out of range
    | assert False
    | Falsifying example: test_format_tracking_label_safe(
    |     s='',
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "/Users/aadhavan/Documents/coll/software/lab/Lab_07/test_processor.py", line 16, in test_format_tracking_label_safe
    |     format_tracking_label(s)
    |   File "/Users/aadhavan/Documents/coll/software/lab/Lab_07/processor.py", line 11, in format_tracking_label
    |     prefix = raw_code[0].upper()
    | TypeError: 'NoneType' object is not subscriptable
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/aadhavan/Documents/coll/software/lab/Lab_07/test_processor.py", line 18, in test_format_tracking_label_safe
    |     assert False, f"format_tracking_label crashed with: {type(e).__name__}: {e}"
    | AssertionError: format_tracking_label crashed with: TypeError: 'NoneType' object is not subscriptable
    | assert False
    | Falsifying example: test_format_tracking_label_safe(
    |     s=None,
    | )
    +------------------------------------
```

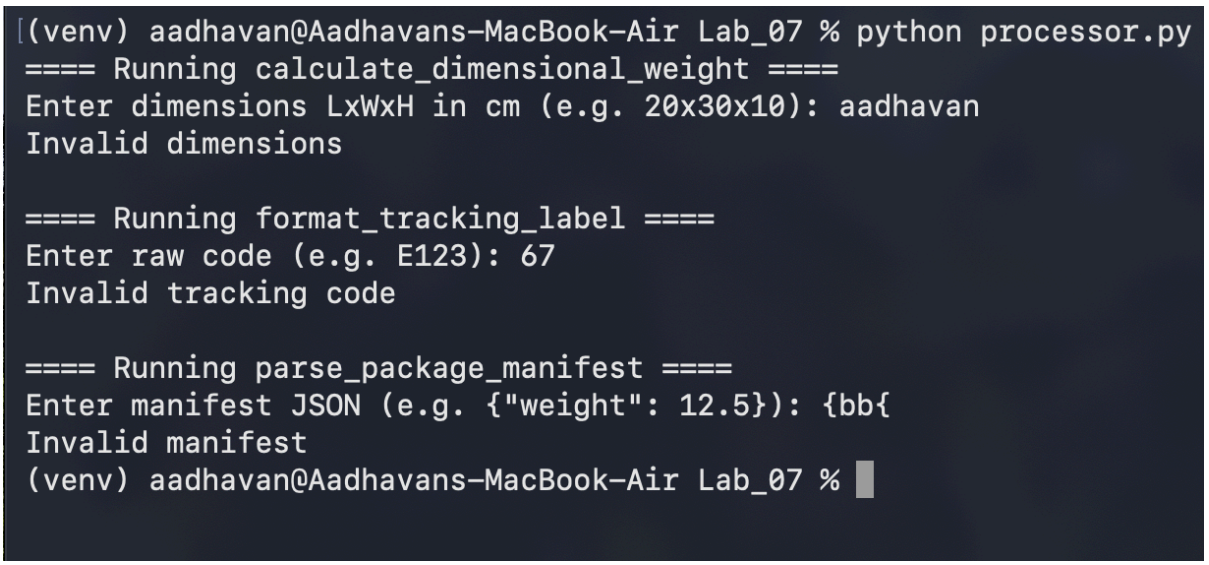


```
_____ test_parse_package_manifest_safe _____
  + Exception Group Traceback (most recent call last):
  |   File "/Users/aadhavan/Documents/coll/software/lab/Lab_07/test_processor.py", line 22, in test_parse_package_manifest_safe
  |     def test_parse_package_manifest_safe(s):
  |   File "/Users/aadhavan/Documents/coll/software/lab/Lab_07/venv/lib/python3.9/site-packages/hypothesis/core.py", line 2124, in wrapped_test
  |     raise the_error_hypothesis_found
  | exceptiongroup.ExceptionGroup: Hypothesis found 4 distinct failures. (4 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/aadhavan/Documents/coll/software/lab/Lab_07/test_processor.py", line 24, in test_parse_package_manifest_safe
    |     parse_package_manifest(s)
    |   File "/Users/aadhavan/Documents/coll/software/lab/Lab_07/processor.py", line 16, in parse_package_manifest
    |     data = json.loads(json_payload)
    |   File "/Library/Developer/CommandLineTools/Library/Frameworks/Python3.framework/Versions/3.9/lib/python3.9/json/__init__.py", line 346, in loads
    |     return _default_decoder.decode(s)
    |   File "/Library/Developer/CommandLineTools/Library/Frameworks/Python3.framework/Versions/3.9/lib/python3.9/json/decoder.py", line 340, in decode
    |     raise JSONDecodeError("Extra data", s, end)
    | json.decoder.JSONDecodeError: Extra data: line 1 column 2 (char 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/aadhavan/Documents/coll/software/lab/Lab_07/test_processor.py", line 26, in test_parse_package_manifest_safe
    |     assert False, f"parse_package_manifest crashed with: {type(e).__name__}: {e}"
    | AssertionError: parse_package_manifest crashed with: JSONDecodeError: Extra data: line 1 column 2 (char 1)
    | assert False
    | Falsifying example: test_parse_package_manifest_safe(
    |     s='00',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/aadhavan/Documents/coll/software/lab/Lab_07/test_processor.py", line 24, in test_parse_package_manifest_safe
    |     parse_package_manifest(s)
    |   File "/Users/aadhavan/Documents/coll/software/lab/Lab_07/processor.py", line 17, in parse_package_manifest
    |     return float(data["weight"])
    | TypeError: 'int' object is not subscriptable
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/aadhavan/Documents/coll/software/lab/Lab_07/test_processor.py", line 26, in test_parse_package_manifest_safe
    |     assert False, f"parse_package_manifest crashed with: {type(e).__name__}: {e}"
    | AssertionError: parse_package_manifest crashed with: TypeError: 'int' object is not subscriptable
    | assert False
    | Falsifying example: test_parse_package_manifest_safe(
    |     s='0',
    | )
```



Fixed Output

Running python processor.py after the fix: invalid inputs now print error messages instead of crashing.



Passing Tests

Running python -m pytest test_processor.py against the fixed processor.py: all 3 tests pass.

